Embedded Memory Design Phase 1

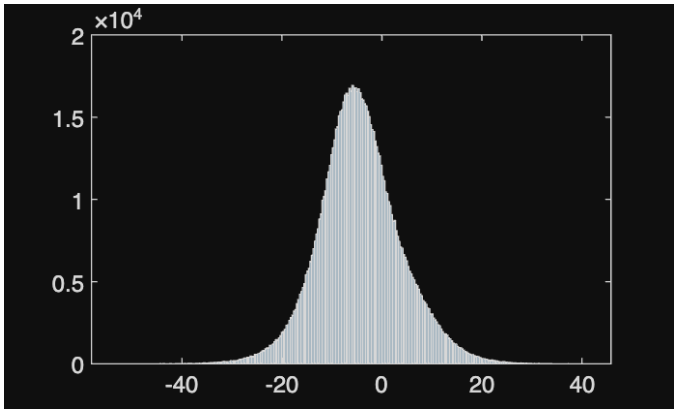
MATLAB Analysis

1. B_x (unsigned)
   Total number of bits=2
   Fractional bits=2

   B_w (signed)
   Total number of bits=6
   Fractional bits=3

Swept through a range of values to find minimum Bx and Bw that meets the required spec for accuracy.

Output histogram before ADC quantization and clipping:



Sweeping through a range of clip points for [-R, R] and ADC bits, got the minimum spec for R = 15 and 4 bits for ADC.

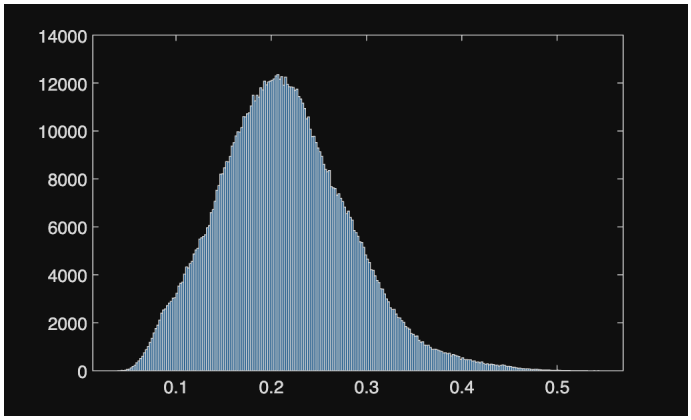
But our architecture uses the idea of Pico Ram paper to shift the weights to positive values and compute the MAC and then digitally add the shifted value. This makes the architecture simple and usable for both negative and positive weighted layers

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From the C-2C paper by Hechen Wang using this architecture for low pvt variations and high linearity focusing on low Imac.

Since this averages out the product our output analog voltage will be averaged values of the actual value. Will need a higher precision ADC for this.

Taking these into consideration and from matlab code representing this flow, the plot for averaged MAC output looks like



Clipping at 0.45 and for ADC precision for 7 bits the accuracy meets 96.78%.

Electric and LTSpice Circuit Design and simulation

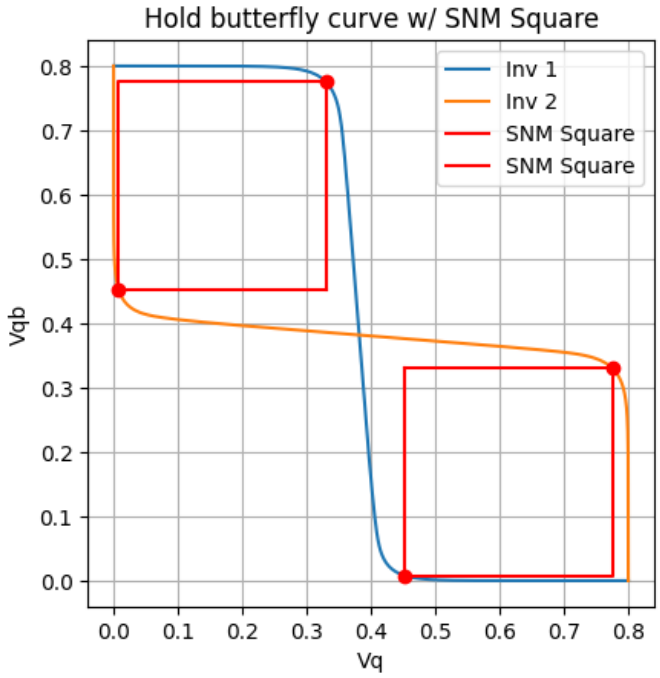
1. **Choice of compute domain:**
   Charged domain
   - Since MAC operation is done through charge sharing, current is not drawn throughout the operation lowering I_{MAC}
   - Lower PVT variations
   - Improved linearity when compared with Current and Time domains

2. **Choice of compute cell**
   9T cell
   6T SRAM cell + 3T for distrubance isolation

   The 3T consisting of an inverter and access transistor to CBL(Compute Bit-Line) provide isolation of the cell ensuring noise picked up on the CBl doesn't cause cell disturb.

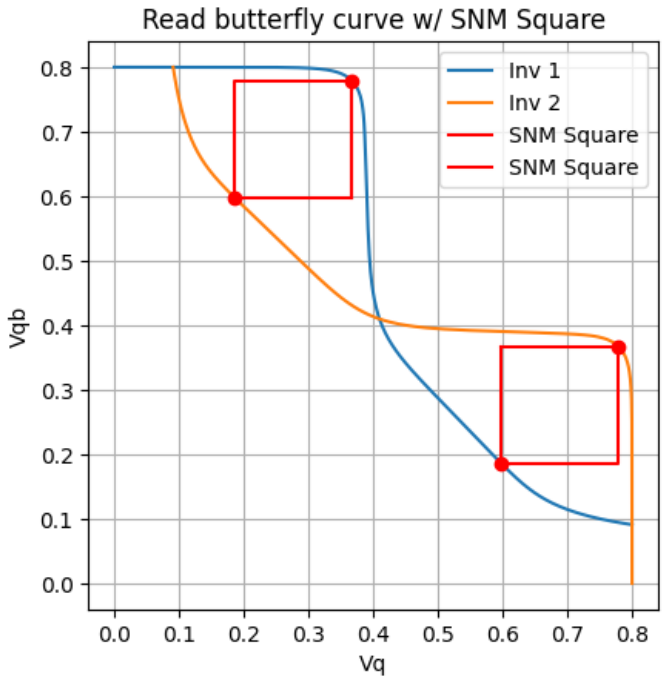
3. **Butterfly curves simulation and margin values for the 6T+3T SRAM cell**

   Hold SNM:



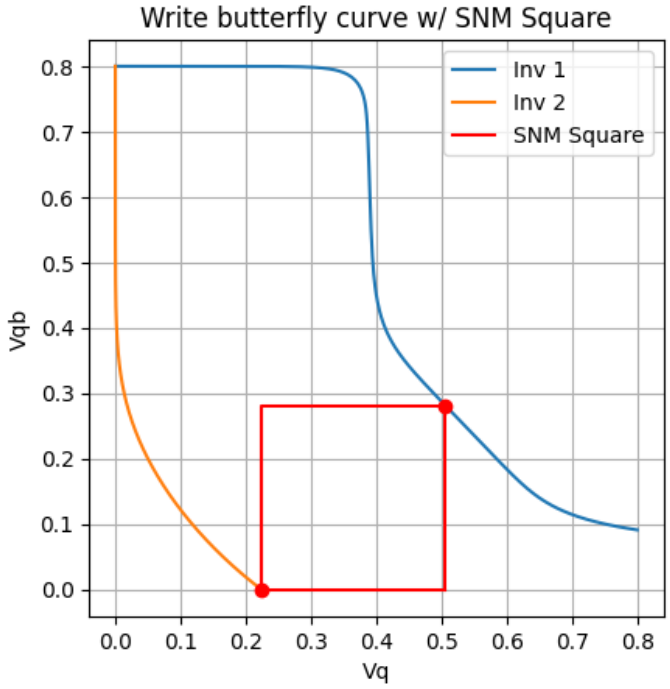
   Diagonal of largest fitted square = 0.456V

Read SNM:



Diagonal of largest fitted square = 0.256V

Write SNM:



Diagonal of largest fitted square = 0.407V

4. **Input D-to-A converter design:**
   R2R Ladder with resistances 1kOhm and 2kOhm was used as DAC to convert 2-bit input signal into analog value with levels 0,0.2V, 0.4V and 0.6V
   The resistor value was chosen to reduce current consumption by DAC while maintaining reasonable fluctuations. Fluctuations reduced while current consumption increased with lower resistance.
   R2R ladder has the drive power to drive the analog input across many rows of SRAM cells

5. **Memory Organisation**
   R=784
   C=600
   N_{ADC}=100

   A single column has 6 parallel 9T cells for 6 bit weight and 784 rows. There is 1 ADC for every column. The output of each row is summed and averaged before sending to ADC

6. **Single column simulation**
   Simulation for a single weight vector with 6 bits and analog input was performed to demonstrate the multiplication operation.
   A C2C ladder is used for multibit weight multiplication.
   Each branch is binary weighted, with output reflected by the status of the switches controlled by CBL.
   As compared to binary weighted capacitors, a C2C ladder uses fewer unit capacitors.

   Output voltage was measured for weight values from 0 to 2^6-1 and plotted to obtain a linear graph.

